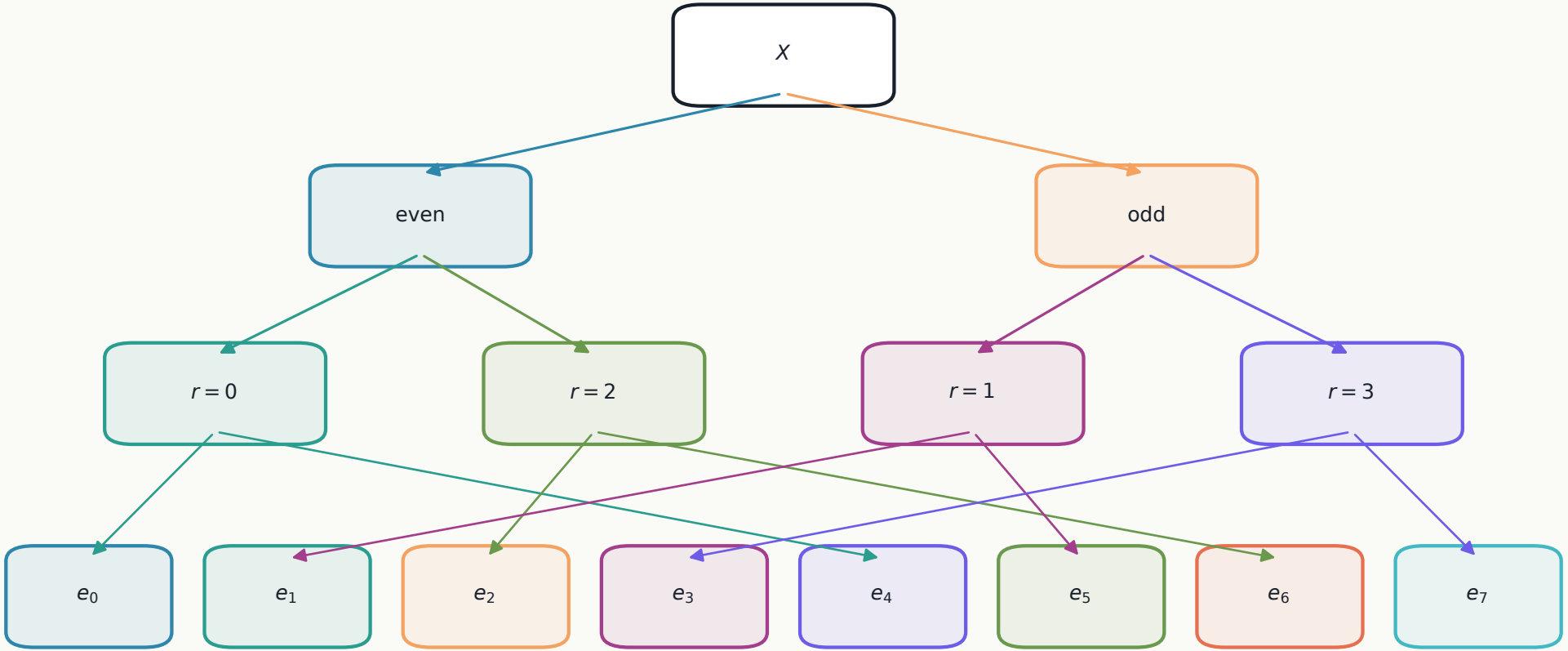
Address refinement: parity -> residue mod 4 -> individual coordinates



$P_{\text{even}} = R_0 + R_2$, $P_{\text{odd}} = R_1 + R_3$, and $R_r = E_r + E_{r+4}$.